EXPERIMENT 15

Probabilistic Context-Free Grammar (PCFG) Parsing

AIM

To implement Probabilistic Context-Free Grammar (PCFG) parsing in Python and compute derivation probabilities using the Viterbi algorithm.

PROCEDURE

1. Define a PCFG specifying production probabilities $\sum P(A \to \alpha) = 1.0$.
2. Instantiate the `ViterbiParser` with the PCFG rules.
3. Parse an ambiguous sentence capable of multiple syntactic structures.
4. Compute the probability score $P(Tree) = \prod P(Rule)$ for each derivation.
5. Output the most probable parse tree and its corresponding probability.

PROGRAM

```
import nltk
from nltk import PCFG
from nltk.parse import ViterbiParser

pcfg_grammar = PCFG.fromstring("""
    S -> NP VP           [1.0]
    VP -> V NP           [0.7] | VP PP [0.3]
    PP -> P NP           [1.0]
    NP -> Det N          [0.6] | NP PP [0.2] | 'astronomers' [0.2]
    Det -> 'the'         [0.6] | 'a'   [0.4]
    N -> 'stars'         [0.5] | 'telescope' [0.5]
    V -> 'saw'           [1.0]
    P -> 'with'          [1.0]
""")

viterbi = ViterbiParser(pcfg_grammar)
sentence = "astronomers saw the stars with a telescope".split()

print(f"Input Sentence: {' '.join(sentence)}")
print("=" * 60)

parses = list(viterbi.parse(sentence))

if parses:
    best_tree = parses[0]
    print(f"Most Likely Parse Tree (Viterbi Optimal):")
    print(best_tree)
    print(f"\nParse Log Probability: {best_tree.prob():.6e}")
else:
    print("No parse found.")
```

OUTPUT

```
> python3 experiment_15.py
------------------------------------------------------------------------------------------
Input Sentence: astronomers saw the stars with a telescope
============================================================
Most Likely Parse Tree (Viterbi Optimal):
(S
  (NP astronomers)
  (VP
    (VP (V saw) (NP (Det the) (N stars)))
    (PP (P with) (NP (Det a) (N telescope))))) (p=0.003024)

Parse Log Probability: 3.024000e-03
[Process completed with exit code 0]
```

RESULT

Thus, PCFG parsing using the Viterbi algorithm was successfully implemented, finding the highest-probability syntactic derivation.